



Lab Number : 10

Date:2025/08/17

Title: Implementation of Dynamic Interior/ Exterior Routing (RIP, OSPF, BGP)

THEORY:

Dynamic Interior Routing

Dynamic Interior Routing deals with the automatic exchange and updating of routes within a single Autonomous System (AS). It enables routers inside the same organization or network to share routing information and determine the most efficient path to reach a destination. Commonly used interior routing protocols include RIP (Routing Information Protocol), OSPF (Open Shortest Path First), and EIGRP (Enhanced Interior Gateway Routing Protocol).

Dynamic Exterior Routing

Dynamic Exterior Routing, on the other hand, manages the flow of routing information between different Autonomous Systems (ASes). It allows communication across organizational or network boundaries, which is crucial for the functioning of the Internet. The primary protocol used here is BGP (Border Gateway Protocol), the standard exterior routing protocol. BGP maintains route information between ASes, ensuring reliable and optimized data exchange across global networks.

RIP (Routing Information Protocol) is a distance-vector routing protocol that determines the best route by counting the number of hops. Since it has a maximum limit of 15 hops, it is mainly useful in small-scale networks. The protocol is simple to configure, but it suffers from slow convergence and does not scale well, making it less effective for large or complex network environments.

OSPF (Open Shortest Path First) is a link-state routing protocol that relies on the Dijkstra algorithm to calculate the most efficient path. It is ideal for larger networks because it can be organized into multiple areas for better management. OSPF provides quick convergence, supports features such as route summarization and variable-length subnet masks (VLSM), but compared to RIP, it is more complex to set up and maintain.

BGP (Border Gateway Protocol) is a path-vector routing protocol designed to handle routing between different Autonomous Systems (ASes), such as Internet Service

Providers and large organizations. Unlike interior protocols, BGP emphasizes policy-based routing, allowing administrators to control how traffic flows across the Internet. It is extremely scalable and essential for global connectivity, but it has slower convergence and requires careful, more complex configuration compared to protocols like OSPF.

Network Diagram

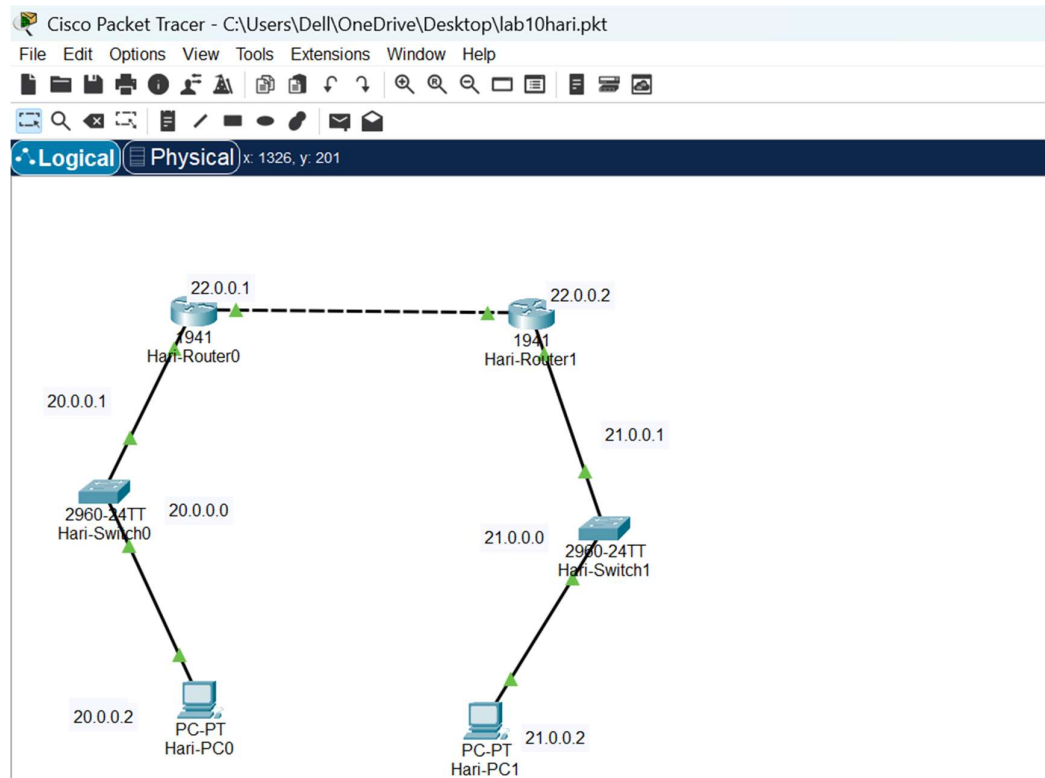


Fig: Network Diagram

Implementation Sequence

Here is the implementation sequence for Implementation of Dynamic Interior/ Exterior Routing (RIP, OSPF, BGP).

1. Configuring Network

Configure the network for PCs and Routers

Step 1: Open Packet Tracer and setup the devices.

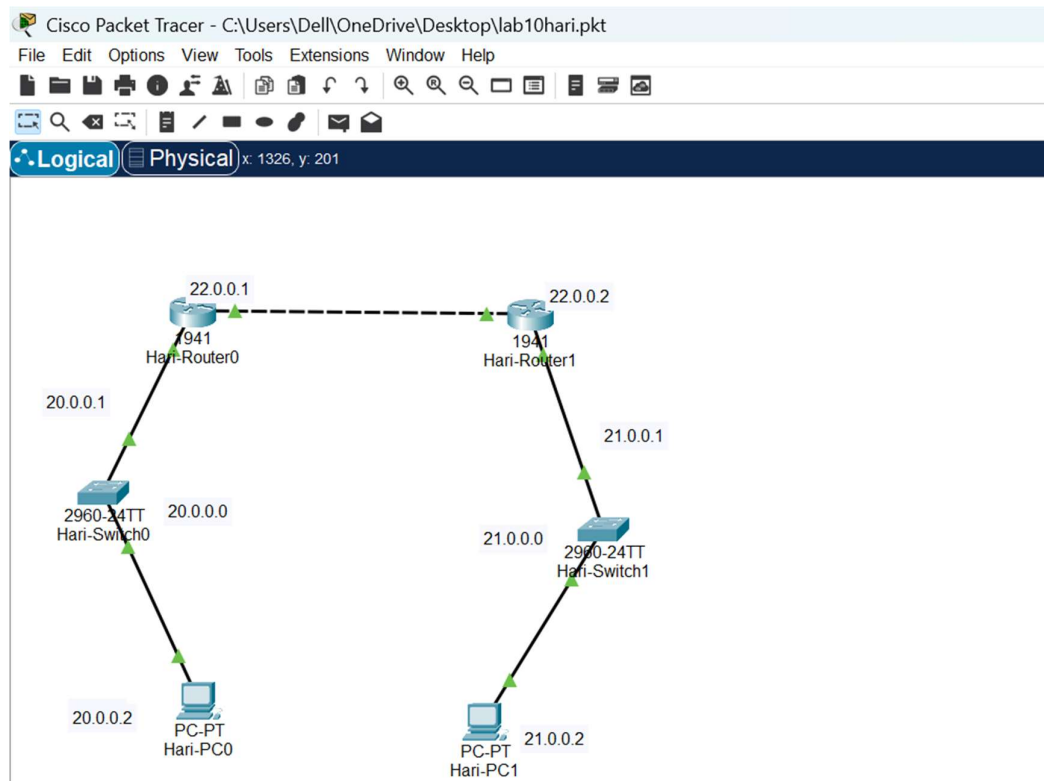
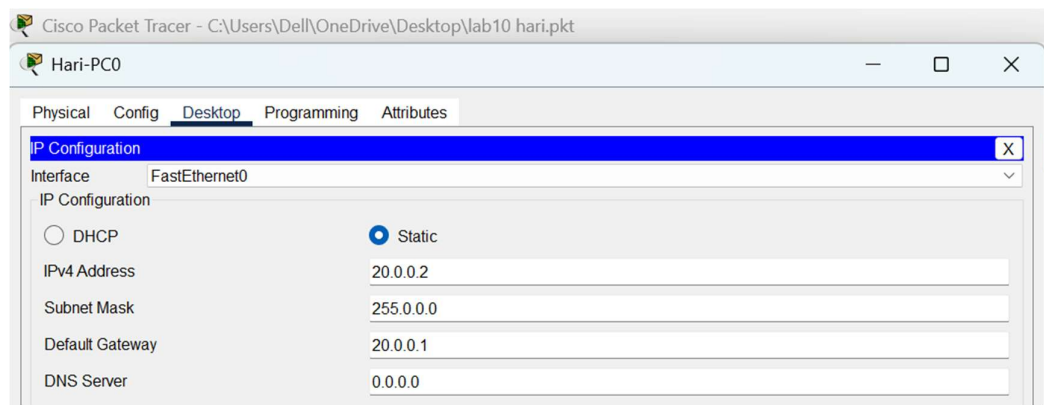


Fig: Simple Network setup

Step 2: Assign IP addresses to each PC.



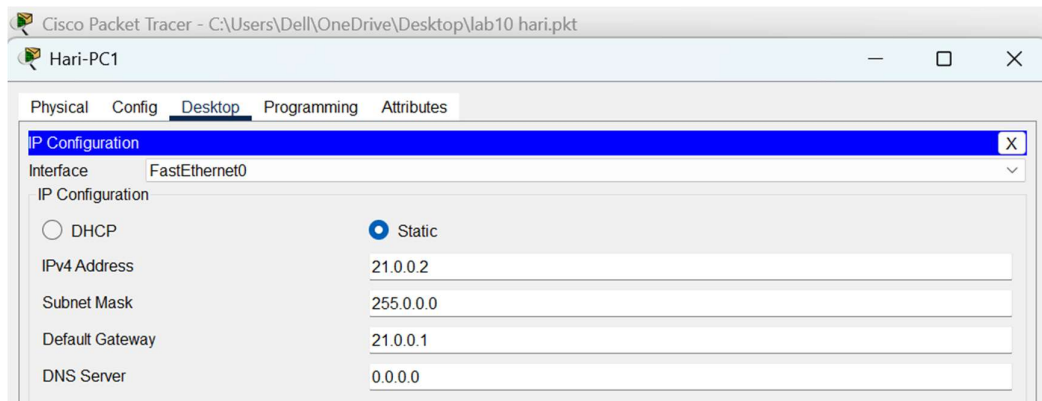


Fig: IP configuration

Step 3: Assign IP addresses to router interfaces.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 22.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface GigabitEthernet 0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

Fig: Gigabit Ethernet configuration on router0

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 22.0.0.2 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 21.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

Fig: Gigabit Ethernet configuration on router1

Implementation & explain the need for Dynamic Routing

Dynamic routing protocols such as **RIP, OSPF, and BGP** allow routers to automatically learn and update routes as the network changes. The general steps for implementation are:

1. Network Configuration

- Set up the network topology by connecting routers, switches, and PCs.
- Assign appropriate IP addresses and subnet masks to each interface.

2. Dynamic Routing Setup

- Enable the required dynamic routing protocol (e.g., RIP for small networks, OSPF for larger enterprise networks, or BGP for inter-AS communication).
- Configure the routers so they can exchange routing updates with their neighbors.

3. Testing and Verification

- Use commands such as **ping** to check connectivity between devices.

Need for Dynamic Routing

Dynamic routing plays an essential role in modern networks because of the following benefits:

- **Automatic Route Updates:** When network conditions change—such as link failures or the addition of new devices—dynamic routing protocols automatically refresh routing tables. This ensures uninterrupted connectivity without manual intervention.
- **Scalability:** For large or frequently changing networks, manually configuring static routes becomes difficult. Dynamic routing protocols manage network growth more effectively, making expansion easier.
- **Efficient Path Selection:** These protocols continuously analyze network conditions to select the most optimal path for data transmission, improving overall performance and reducing latency.
- **Redundancy and Fault Tolerance:** In the event of failures, dynamic routing quickly adapts and redirects traffic through alternative routes, maintaining reliability and minimizing downtime.

- **Load Balancing:** Many dynamic routing protocols can spread traffic across multiple available paths, preventing congestion on a single route. This enhances both efficiency and network performance.

Dynamic Routing Configuration

1. Using RIP Command

Network Diagram

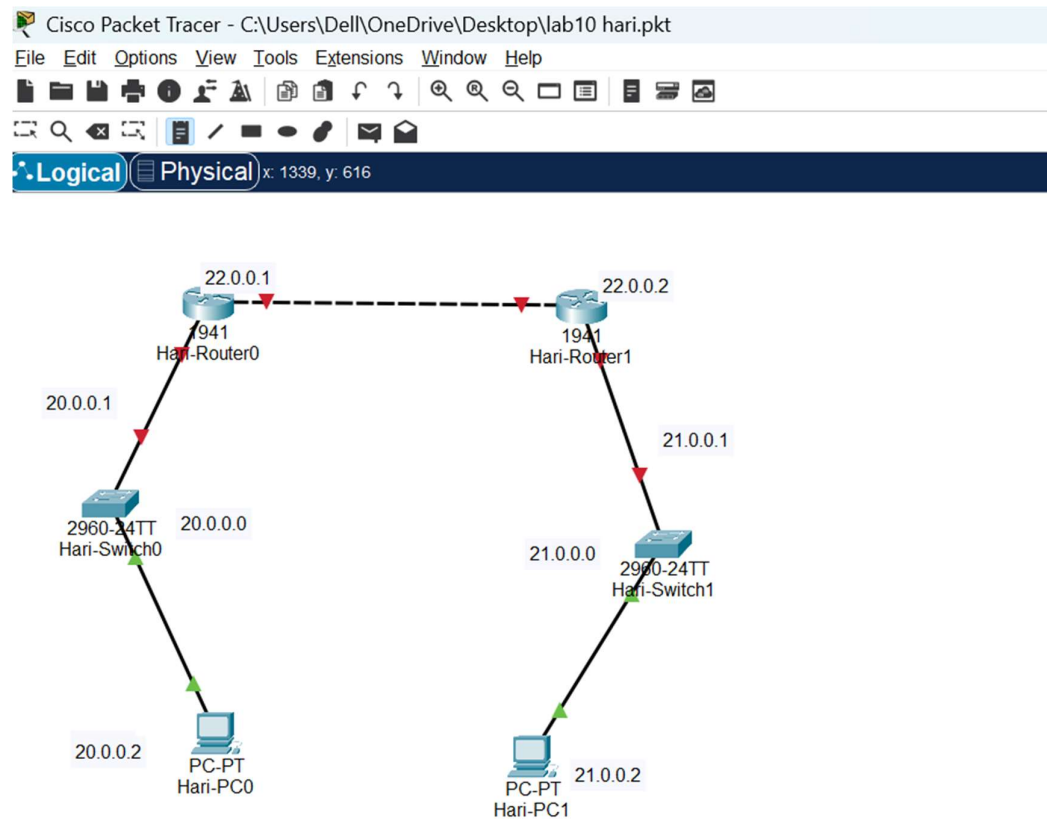


Fig: Network Diagram

Code for Dynamic Routing Configuration using RIP command

For Router 0

Router>enable

Router#configure terminal

Router(config)#router rip

Router(config-router)#version 2

```
Router(config-router)#network 22.0.0.0
```

```
Router(config-router)#network 21.0.0.0
```

```
Router(config-router)#exit
```

For Router 1

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#router rip
```

```
Router(config-router)#version 2
```

```
Router(config-router)#network 20.0.0.0
```

```
Router(config-router)#network 22.0.0.0
```

```
Router(config-router)#exit
```

Steps for Dynamic Routing Configuration using RIP Command

Step 1: Access the Router CLI

- Open Packet Tracer
- Click on the router we want to configure to access its CLI (Command Line Interface).

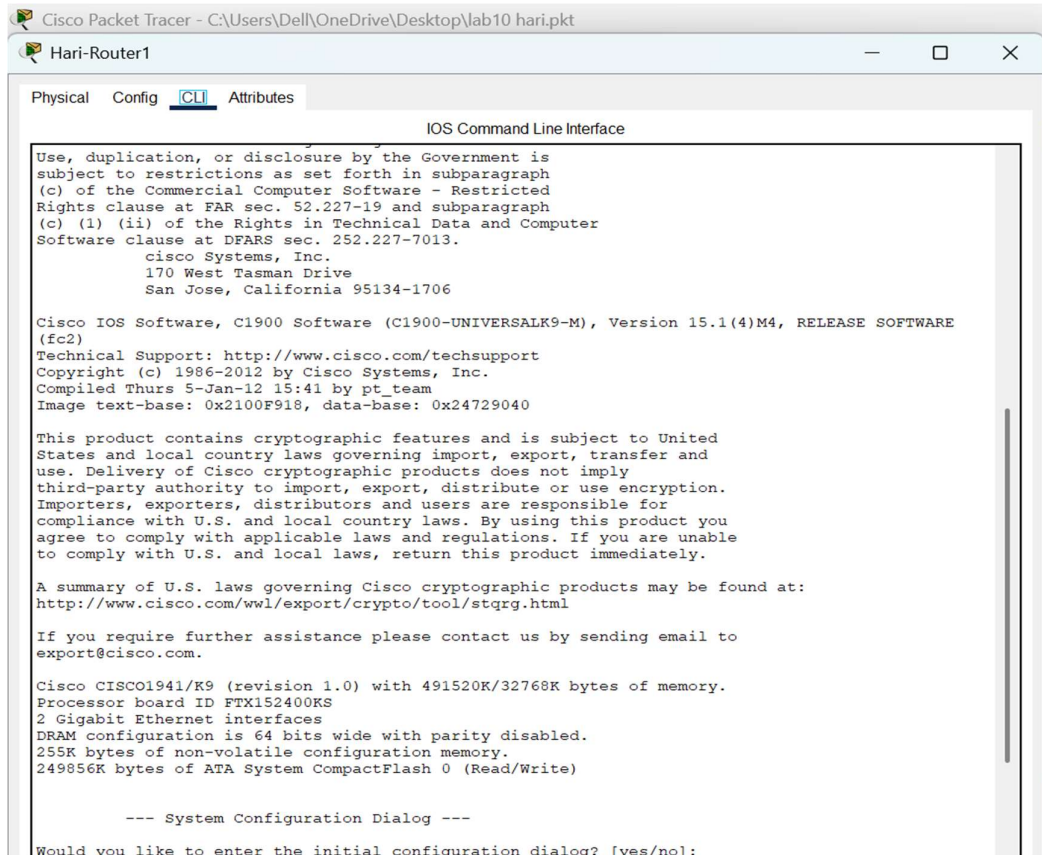


Fig: Accessing Router CLI via Console Cable

Step 2: Enter Global Configuration Mode

- Enable RIP Routing Protocol
- Specify the RIP Version
- Configure the RIP Networks
- Exit RIP Configuration Mode

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 22.0.0.0
Router(config-router)#network 21.0.0.0
Router(config-router)#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 20.0.0.0
Router(config-router)#network 22.0.0.0
Router(config-router)#
```

Fig: Router configuration of router0 and router1 using RIP command

Testing

To test whether the network is working, we can ping other devices on the network from each PC. If the ping is successful, we should see replies from the other device.

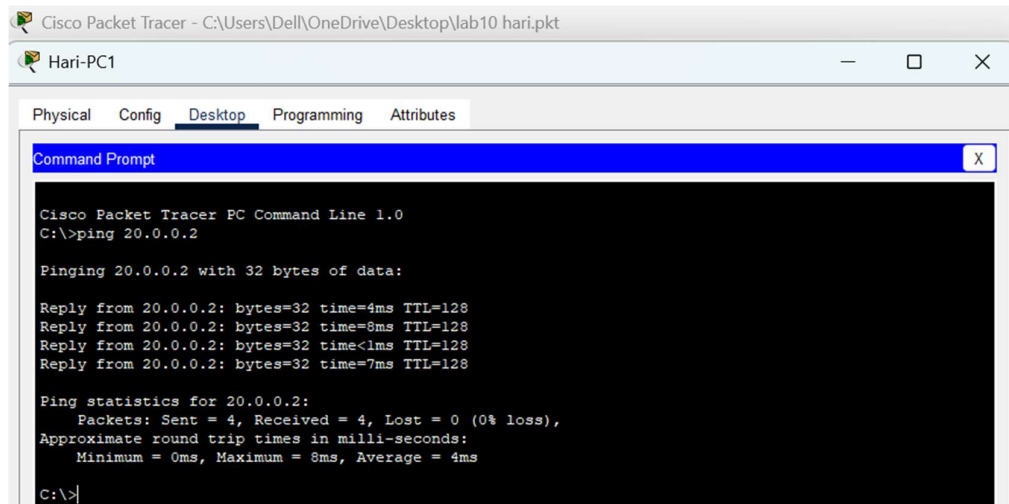


Fig: Connectivity test between Hari-PC1 to Hari-PC0

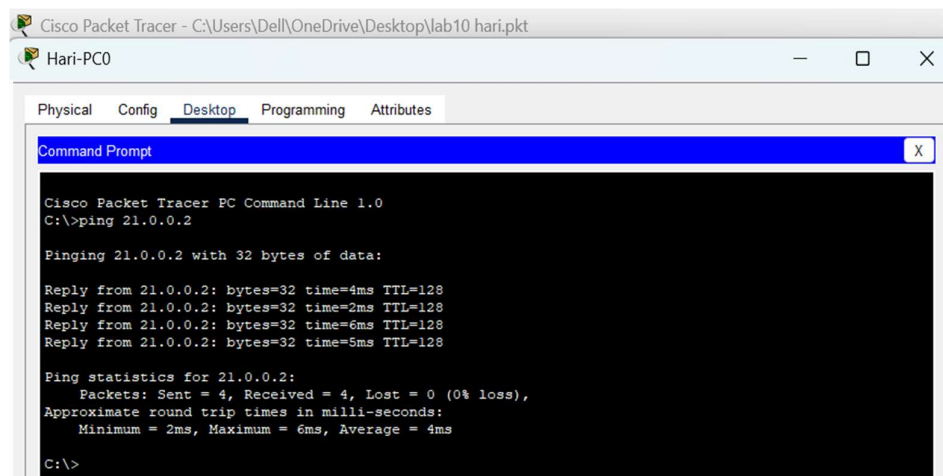


Fig: Connectivity test between Hari-PC0 to Hari-PC1

Above figure shows the results of a ping test between two PCs on a network. The ping test was successful, indicating that there is connectivity between the two devices. The output of the ping test shows the number of packets sent, received, and lost, as well as the approximate round trip time for each packet.

2.Using OSPF Command

Network Diagram

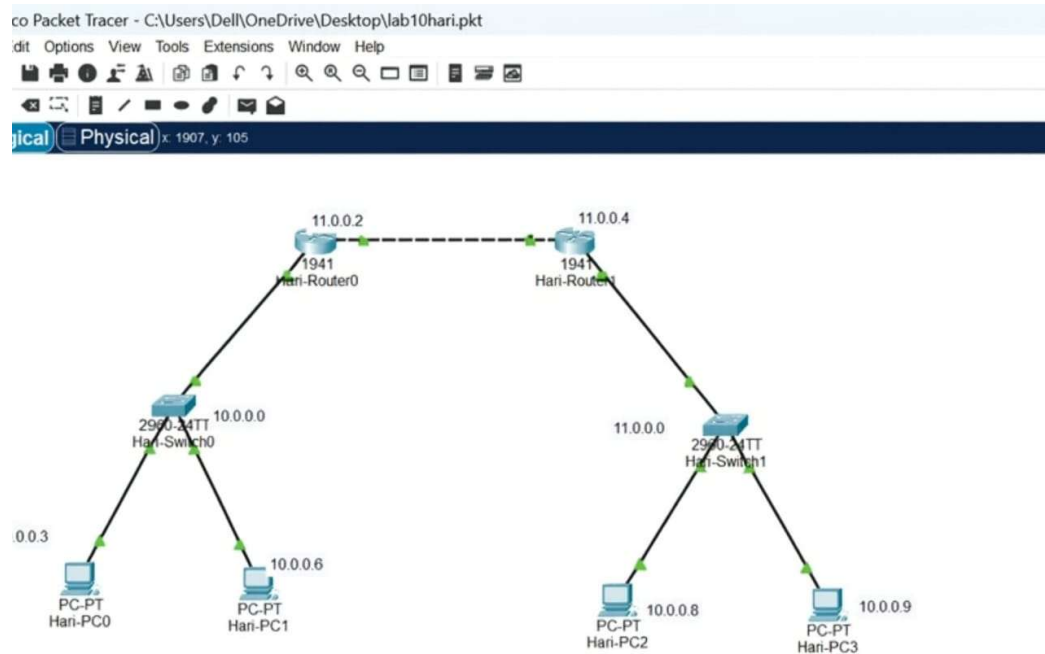


Fig: Network Diagram

Code for Dynamic Routing Configuration Using OSPF Command

For Router0

```
Router>enable
```

```
Router#config t
```

```
Router(config)#router ospf 1
```

```
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
```

```
Router(config-router)#network 10.0.1.0 0.0.0.255 area 0
```

```
Router(config-router)#exit
```

For Router1

```
Router>enable
```

```
Router#config t
```

```
Router(config)#router ospf 2
```

```
Router(config-router)#network 10.0.3.0 0.0.0.255 area 0
```

```
Router(config-router)#network 10.0.1.0 0.0.0.255 area 0
```

Router(config-router)#exit

Steps for Dynamic Routing Configuration using OSPF command

Step 1: Access the Router CLI

- ❖ Open Packet Tracer.
- ❖ Click on the router we want to configure to access its CLI (Command Line Interface).

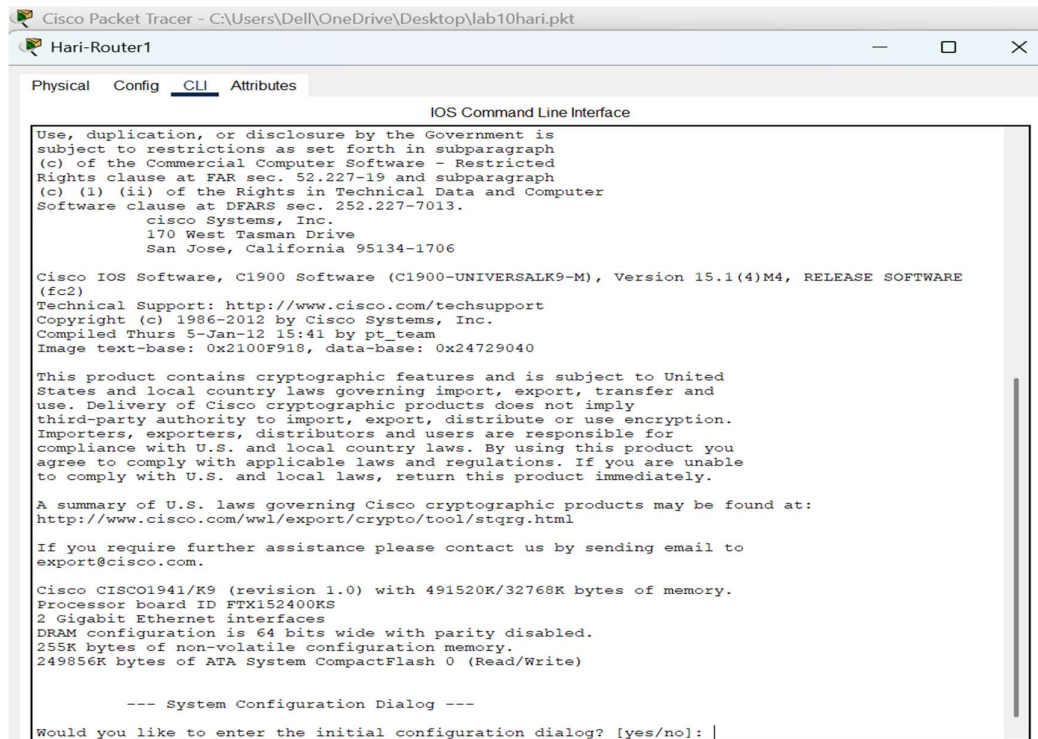


Fig: Accessing Router Command-Line Interface via Console Cable

Step 2: Enter Global Configuration Mode

- Enable OSPF Routing Protocol
- Start the OSPF process and assign it a process ID
- Assign a router ID
- Specify the networks connected to Router 1, and define the areas.
- Exit OSPF configuration.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 10.0.1.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 2
Router(config-router)#network 10.0.3.0 0.0.0.255 area 0
Router(config-router)#network 10.0.1.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

Fig: Router configuration for router0 and router1 using OSPF command

Testing

To test whether the network is working, we can ping other devices on the network from each PC. If the ping is successful, we should see replies from the other device.

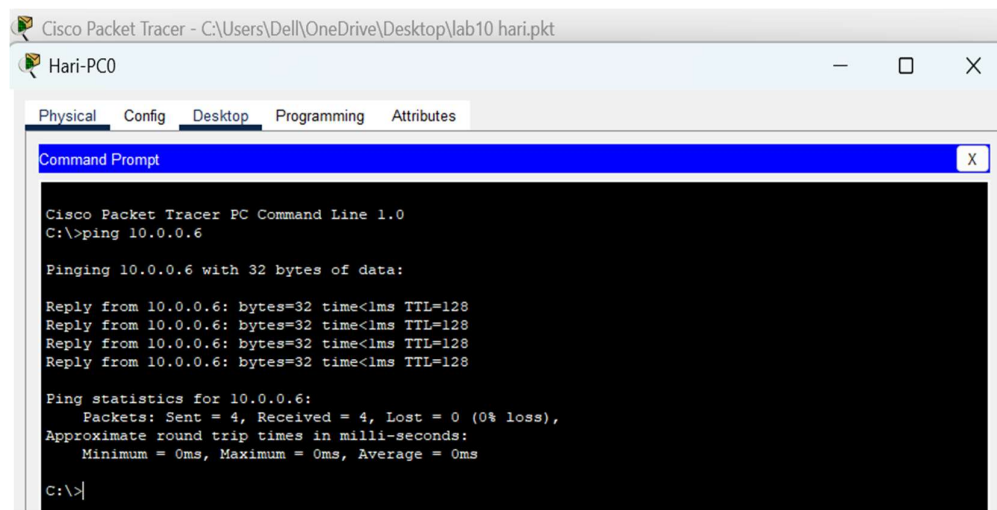


Fig: Connectivity test from Hari-PC0 to Hari-PC1

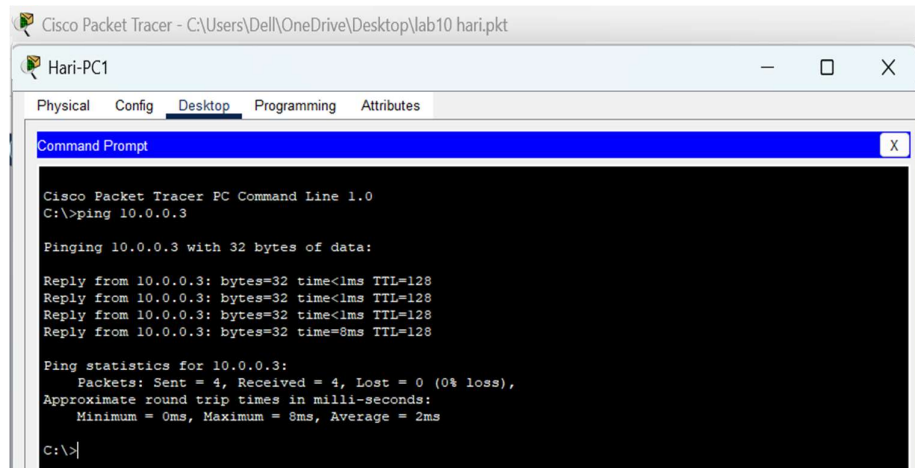


Fig: Connectivity test from Hari-PC1 to Hari-PC0

Above figure shows the results of a ping test between two PCs on a network. The ping test was successful, indicating that there is connectivity between the two devices. The ping output of the ping test shows the number of packets sent, received, and lost, as well as the approximate round trip time for each packet.

Addressing Table:

Device	Interface	IPv4 Address	Subnet	Router no.
PC0	NIC	10.0.0.3	255.0.0.0	0
PC1	NIC	10.0.0.6	255.0.0.0	0
PC2	NIC	10.0.0.8	255.0.0.0	1
PC3	NIC	10.0.0.9	255.0.0.0	1

The addressing table of the OSPF configuration is as follows:

3.Using BGP Command

Network Diagram

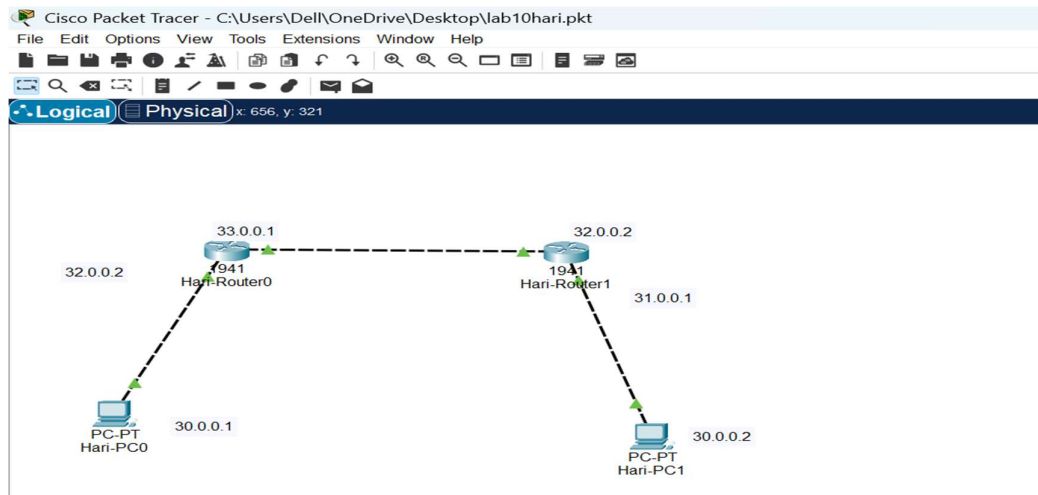


Fig: Network Diagram

Code for Dynamic Routing Configuration Using BGP Command

For Router 0

Router0> enable

Router0# configure terminal

Router(config)# router bgp 420

Router(config-router)# network 199.123.1.0

Router(config-router)# network 199.123.2.0

Router(config-router)# neighbor 199.123.2.2 remote-as 520

Router(config-router)# neighbor 199.123.3.2 remote-as 520

Router(config-router)# exit

For Router1

Router0> enable

Router0# configure terminal

Router(config)# router bgp 420

Router(config-router)# network 199.123.1.0

Router(config-router)# network 199.123.2.0

Router(config-router)# neighbor 199.123.2.1 remote-as 520

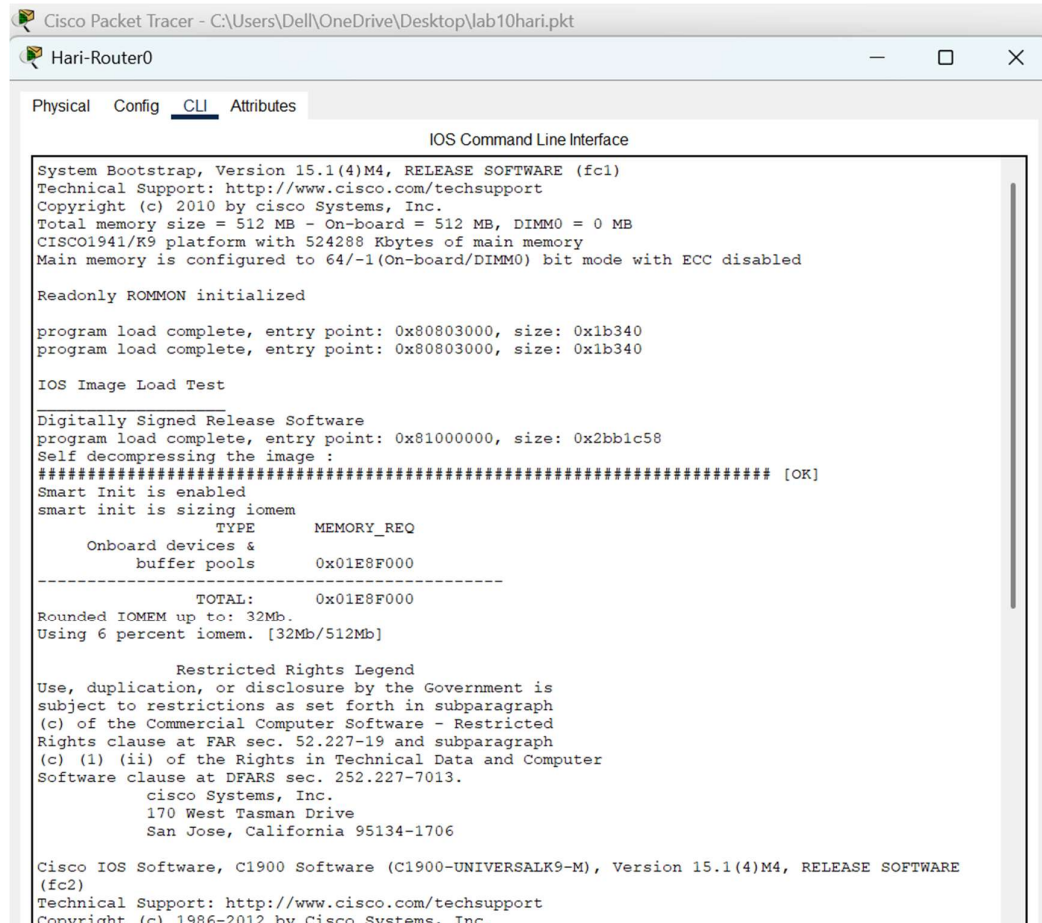
Router(config-router)# neighbor 199.123.1.2 remote-as 520

Router(config-router)# exit

Steps For Dynamic Routing Configuration Using BGP Command

Step 1: Access the Router CLI

- Open Packet Tracer
- Click on the router we want to configure to access its CLI (Command Line Interface).



```
Cisco Packet Tracer - C:\Users\Dell\OneDrive\Desktop\lab10hari.pkt
Hari-Router0
Physical Config CLI Attributes
IOS Command Line Interface
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by Cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
      TYPE      MEMORY_REQ
Onboard devices &
  buffer pools      0x01E8F000
-----
TOTAL:      0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]

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Fig: Accessing Router CLI via Console Cable

Step 2: Enter Global Configuration Mode

- Enable BGP Routing Protocol
- Start the BGP process and specify the AS number
- Configure Neighbor
- Advertise Networks
- Exit BGP configuration


```

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 420
Router(config-router)#network 10.0.1.0
Router(config-router)#network 10.0.2.0
Router(config-router)#neighbour 10.0.2.2 remote-as 520
      ^
% Invalid input detected at '^' marker.

Router(config-router)#neighbor 10.0.2.2 remote-as 520
Router(config-router)#neighbor 10.0.3.2 remote-as 520
Router(config-router)#exit
Router(config)#

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 520
Router(config-router)#network 10.0.2.0
Router(config-router)#network 10.0.3.0
Router(config-router)#neighbor 10.0.2.1 remote-as 420
Router(config-router)#neighbor 10.0.1.1 remote-as 420
Router(config-router)#exit
Router(config)#

```

Fig: Router configuration for router0 and router1 using BGP command

Testing

To test whether the network is working, we can ping other devices on the network from each PC. If the ping is successful, we should see replies from the other device.

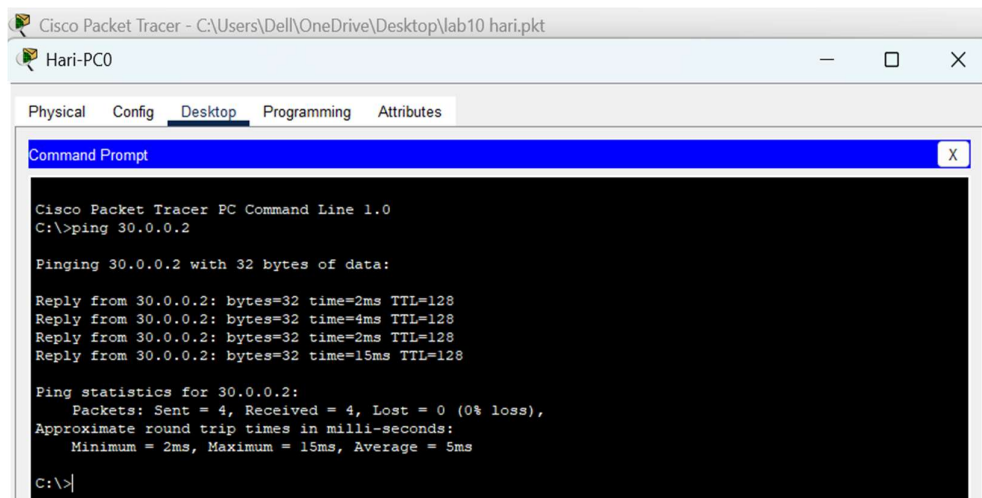


Fig: Connectivity test from Hari-PC0 to Hari-PC1

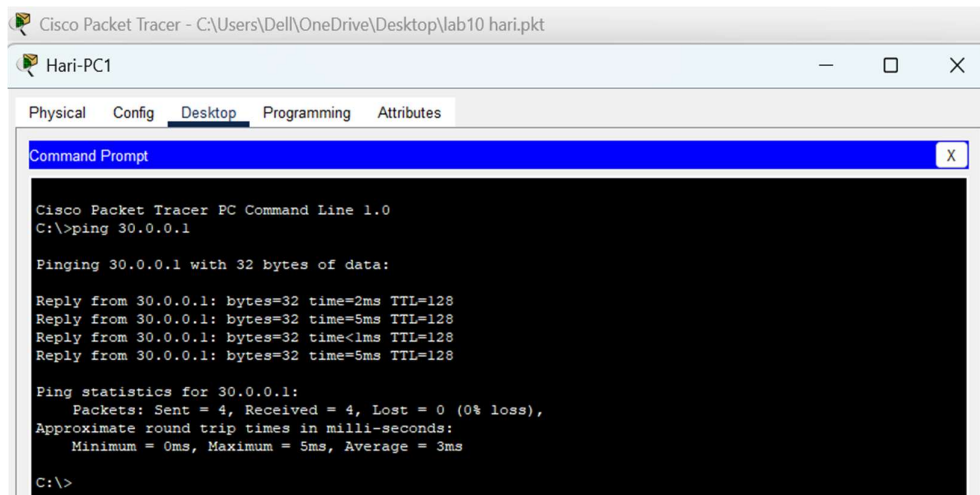


Fig: Connectivity test from Hari-PC1 to Hari-PC0

Above figure shows the results of a ping test between two PCs on a network. The ping test was successful, indicating that there is connectivity between the two devices. The output of the ping test shows the number of packets sent, received, and lost, as well as the approximate round trip time for each packet.

Addressing Table:

The addressing table of the BGP Configuration is as follows:

Device	Interface	IPv4 Address	Subnet	Router No	Link
PC0	NIC	199.123.1.2	255.255.255.0	0	Access
PC1	NIC	199.123.3.2	255.255.255.0	1	Access

CONCLUSION

In this lab, we configured and tested dynamic routing protocols—RIP, OSPF, and BGP to enable automatic route updates and ensure reliable network connectivity. RIP was shown to be simple and effective for smaller networks, while OSPF offered faster convergence and scalability, making it suitable for larger and more complex topologies. BGP demonstrated its importance in exterior routing by managing communication between Autonomous Systems and supporting global Internet traffic. Successful ping tests confirmed correct network setup and validated the smooth functioning of these routing protocols.

